FICTIONAL SAMPLE DOCUMENT FOR WORKSHOP USE. Blue Defence and the content of this document are fictional.

Blue Defence

Sentinel-7 Unmanned Surface Vessel: Maintenance Manual

MM-S7-001 Revision C. Issued 2 February 2026 by the Maritime Systems Division.

1. Vessel overview

Sentinel-7 is a 7.2 m electric unmanned surface vessel for harbour patrol and mine countermeasure trials. This manual covers scheduled maintenance, fastener torques, fault codes and battery safety.

Parameter	Value
Length overall	7.2 m
Maximum speed	25 knots
Endurance	72 hours at 8 knots
Energy storage	4 lithium iron phosphate battery modules, 48 kWh each
Propulsion	2 electric propulsion pods

2. Scheduled maintenance

Record every task in the vessel logbook. Operating hours are read from the vessel management computer.

Component	Task	Interval
Propulsion pod seals	Inspect; replace if worn	Every 250 operating hours
Hull anodes	Replace	Every 6 months, or when 50% consumed
Battery modules	Capacity test	Every 500 operating hours
Radar mast hinge	Lubricate	Every 100 operating hours
Satellite antenna cabling	Inspect for chafing	Every 3 months

3. Fastener torques

Use a calibrated torque wrench. Replace any fastener that shows corrosion.

Fastener	Thread	Torque
Hull access panel bolts	M8	24 N·m
Propulsion pod mounting bolts	M12	85 N·m
Antenna base bolts	M6	10 N·m

4. Fault codes

The vessel management computer reports faults to the shore operator console.

Code	Meaning	Operator action
E-104	Propulsion pod overtemperature	Reduce thrust to 40% and return to base
E-221	GNSS signal lost	Switch to inertial navigation and alert the shore operator
E-305	Battery cell imbalance above 80 mV	Stop charging and run the balancing routine

5. Battery safety

Charge battery modules only when the ambient temperature is between 5 °C and 40 °C.

If a battery module shows smoke, swelling or a temperature above 60 °C, isolate shore power, evacuate everyone to at least 15 m from the vessel, and call the fire service.